Patchy Environments II

Today's Agenda:

- Quiz
- Patchy Environments

Patchy Environments

Patches: suitable habitat, surrounded by unsuitable habitat.

Metapopulation: collection of connected populations

- Multiple patches connected by dispersal

Levins Metapopulation Model

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

$$dP/dt = cP(1 - P) - mP$$

Levins Metapopulation Model and habitat loss

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

D = Proportion of patches destroyed

$$dP/dt = cP(1 - D - P) - mP$$

at $D = 1 - m/c$

Extinction occurs

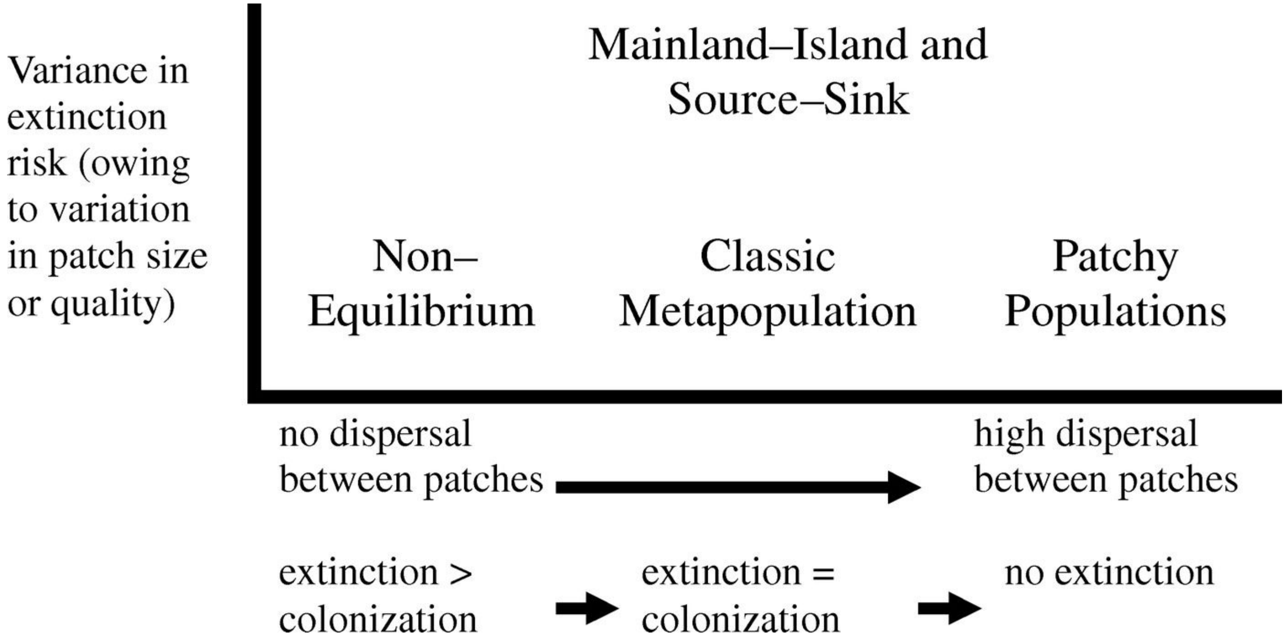
Types of metapopulation structure


1031

REVIEW / SYNTHÈSE

How to find a metapopulation[1]

D.A. Driscoll

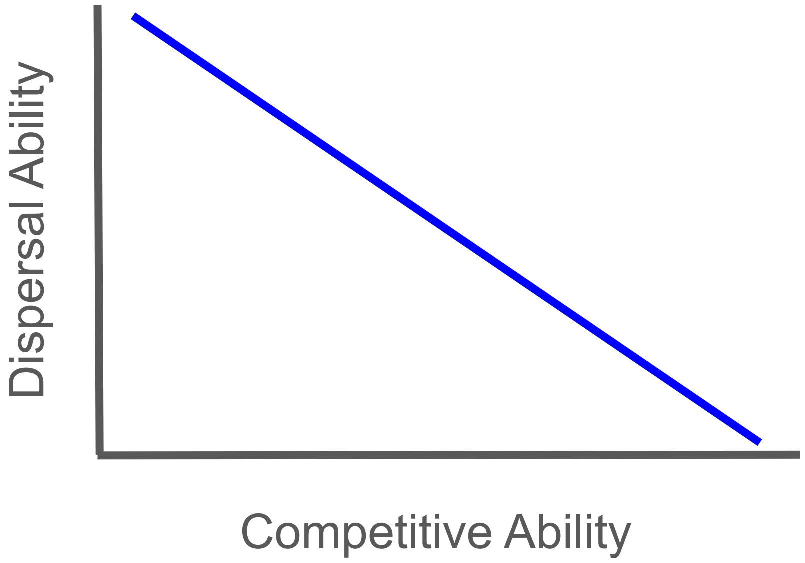


Competition-Colonization Trade-off

For competing species to coexist:

- $a_{ii} > a_{ji}$
- Requires trade-offs

Competition-Colonization Trade-off



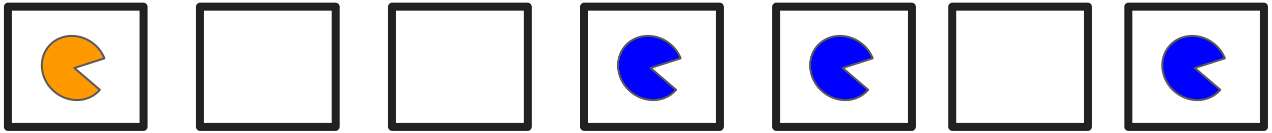
Competition-Colonization Trade-off

Superior colonizer: gets to patches first
Superior competitor: arrives later, but displaces the superior colonizer

Superior colonizer species are concentrated in recently-disturbed patches
Superior competitor species are concentrated in older patches

- Increases a_{ii} relative to a_{ji}

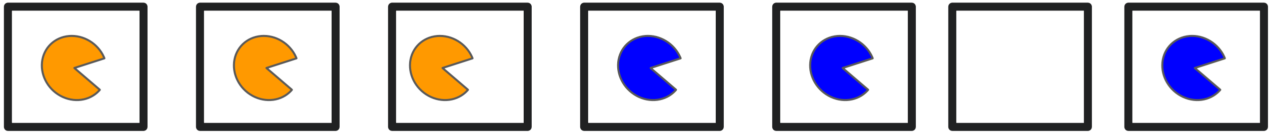
Competition-Colonization Trade-off



 = Better colonizer

 = Better competitor

Competition-Colonization Trade-off



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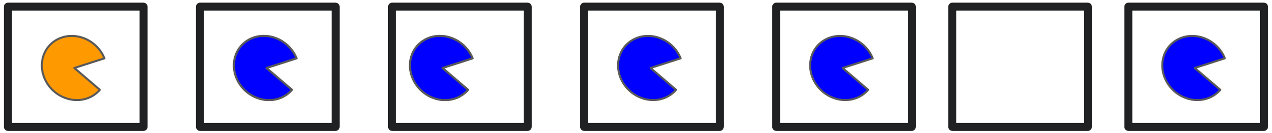
Competition-Colonization Trade-off



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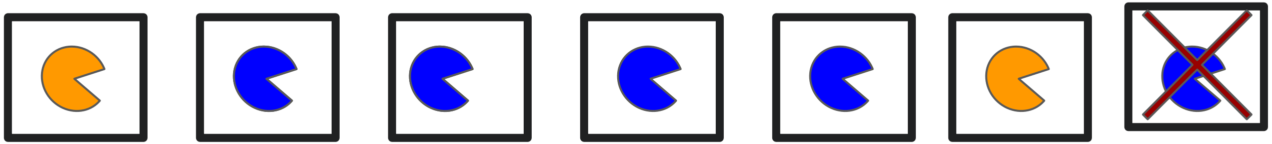
Competition-Colonization Trade-off



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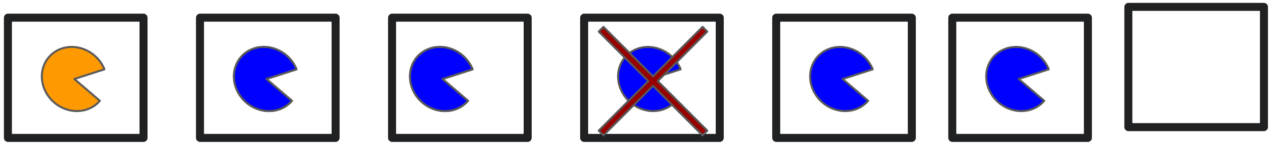
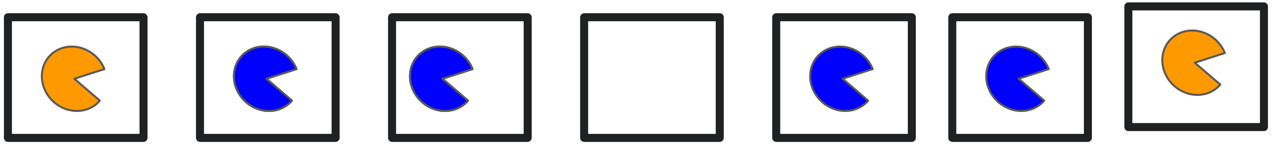
Competition-Colonization Trade-off



 = Better colonizer

 = Better competitor

Competition-Colonization Trade-off



 = Better colonizer

 = Better competitor

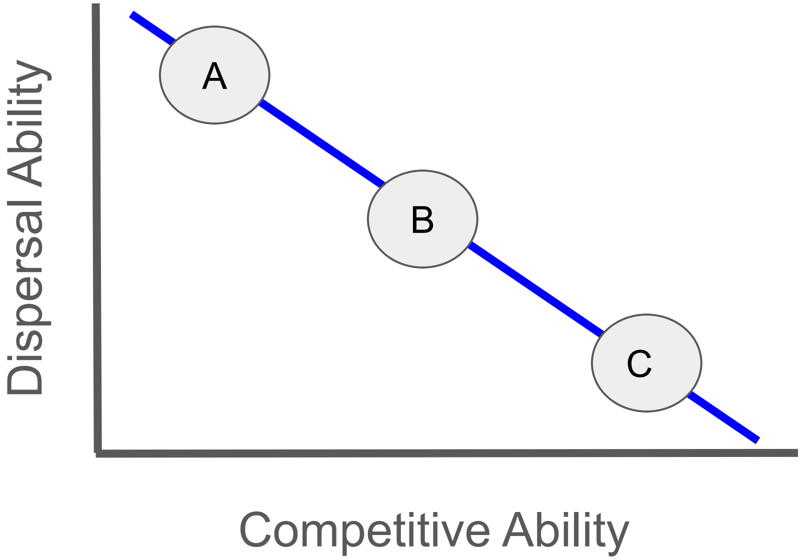
Competition-Colonization Trade-off

 = Better colonizer

 = Better competitor

Competition-Colonization Trade-off

Idea can be generalized to more than two species:



1) A arrives first
2) B arrives next, displaces A
3) C arrives last, displaces B
4) Patch extinction occurs (go to 1)

Competition-Colonization Trade-off

$$dP/dt = cP(1 - P) - mP$$

Competition-Colonization Trade-off

$$dP_i/dt = c_i P_i (1 - P_i) - mP_i$$

$$dP_j/dt = c_j P_j (1 - P_j) - mP_j$$

Competition-Colonization Trade-off

$$dP_i/dt = c_iP_i(1 - P_i) - mP_i$$

$$dP_j/dt = c_jP_j(1 - P_j - P_i) - mP_j - c_iP_iP_j$$

Species i = Better competitor
Species j = Better colonizer

Competition-Colonization Trade-off

$$dP_i/dt = c_i P_i(1 - P_i) - mP_i$$

$$dP_j/dt = c_j P_j(1 - P_j - P_i) - mP_j - c_i P_i P_j$$

Species i = Better competitor
Species j = Better colonizer

To get coexistence conditions:

- Assume equilibrium (dP/dt = 0)
- Both species present ($P_j > 0$)
- Solve for c_i

$$c_j > c_i^2/m$$

Competition-Colonization Trade-off

$$dP_i/dt = c_iP_i(1 - P_i) - mP_i$$

$$dP_j/dt = c_jP_j(1 - P_j - P_i) - mP_j - c_iP_iP_j$$

Species i = Better competitor
Species j = Better colonizer

To get coexistence conditions:

- Assume equilibrium (dP/dt = 0)
- Both species present ($P_j > 0$)
- Solve for c_i

$$c_j > c_i^2/m$$

Higher patch extinction rate, lower c_j needed

- More patches available to colonize

Competition-Colonization: Example

Ecology, 75(1), 1994, pp. 2–16

COMPETITION AND BIODIVERSITY IN SPATIALLY STRUCTURED HABITATS[1]

DAVID TILMAN
Department of Ecology, Evolution, and Behavior, 1987 Upper Buford Circle, University of Minnesota, St. Paul, Minnesota 55108 USA

Tilman looked at this in both experimental and theoretical systems

(Remember the grassland plots we discussed a while back?)

Competition-Colonization: Example

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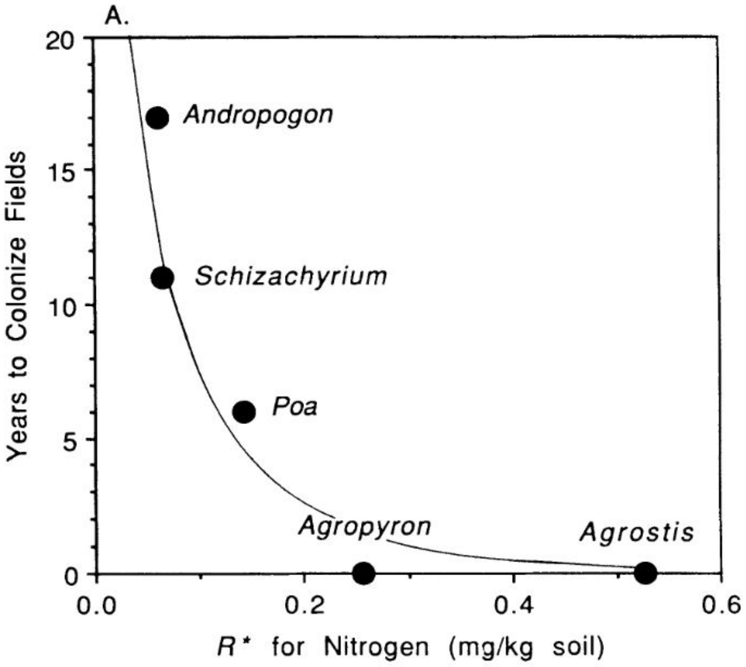

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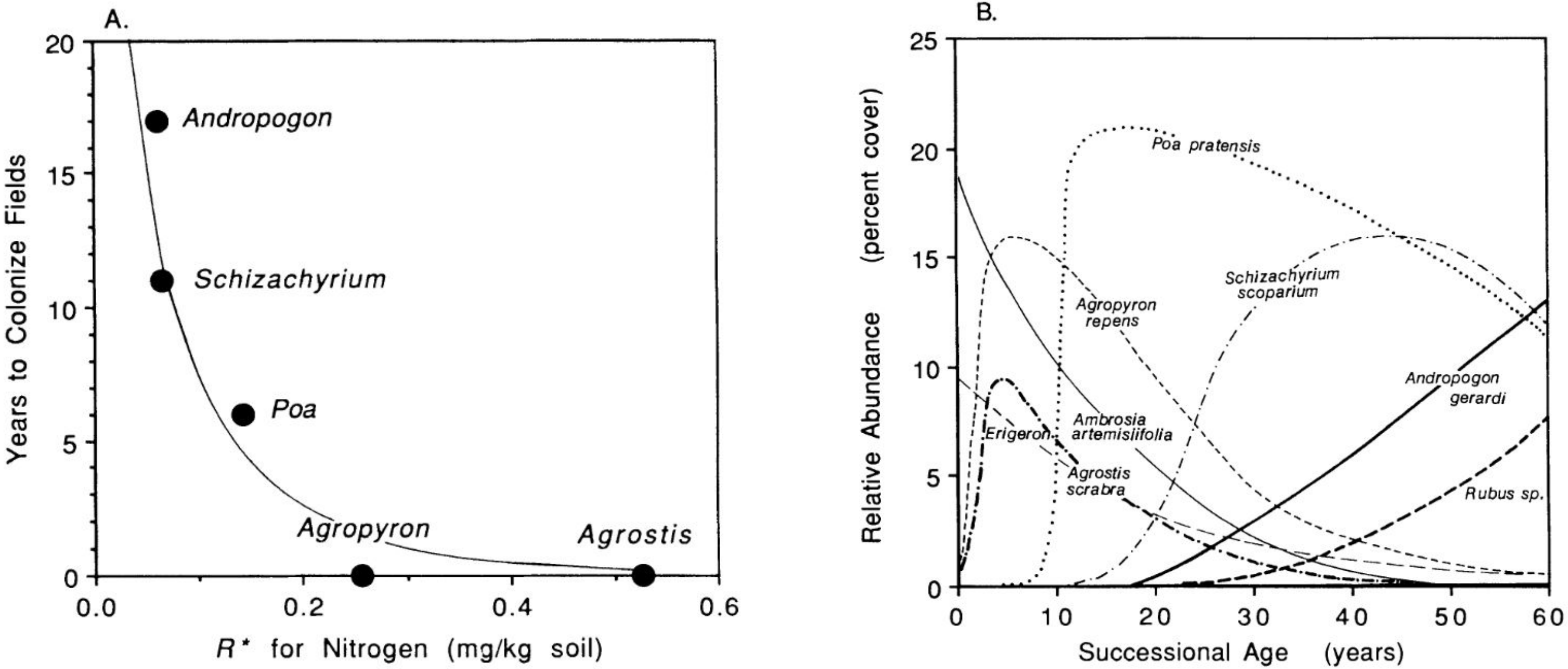
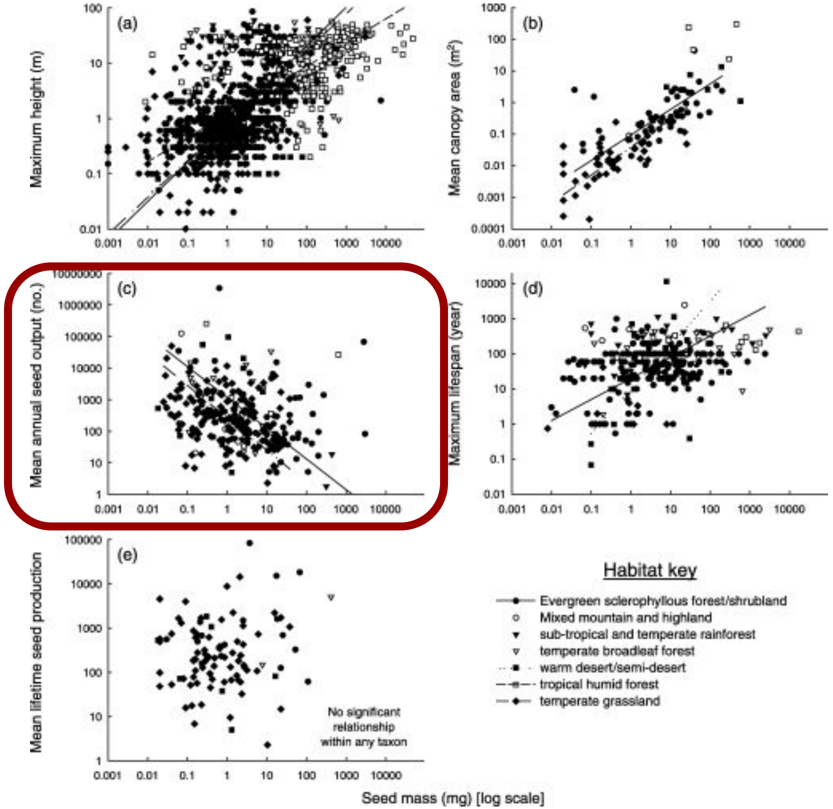


Journal of Ecology 2004
92, 384–396

Small-seeded species produce more seeds per square metre of canopy per year, but not per individual per lifetime

ANGELA T. MOLES*, DANIEL S. FALSTER, MICHELLE R. LEISHMAN and MARK WESTOBY
Department of Biological Sciences, Macquarie University, Sydney, NSW 2109, Australia

Competition-Colonization and Seed Traits



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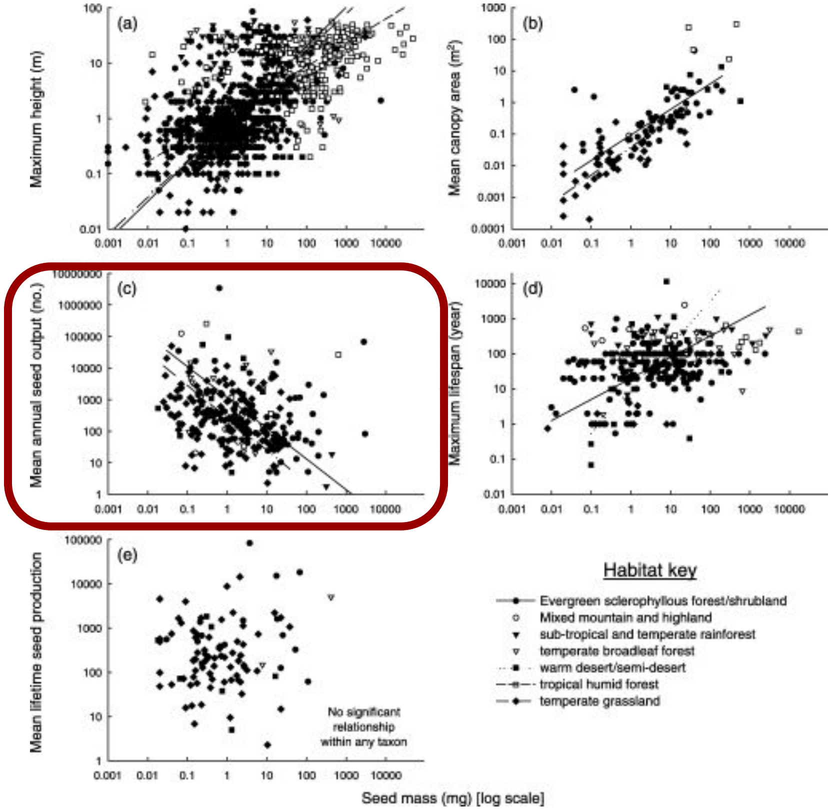
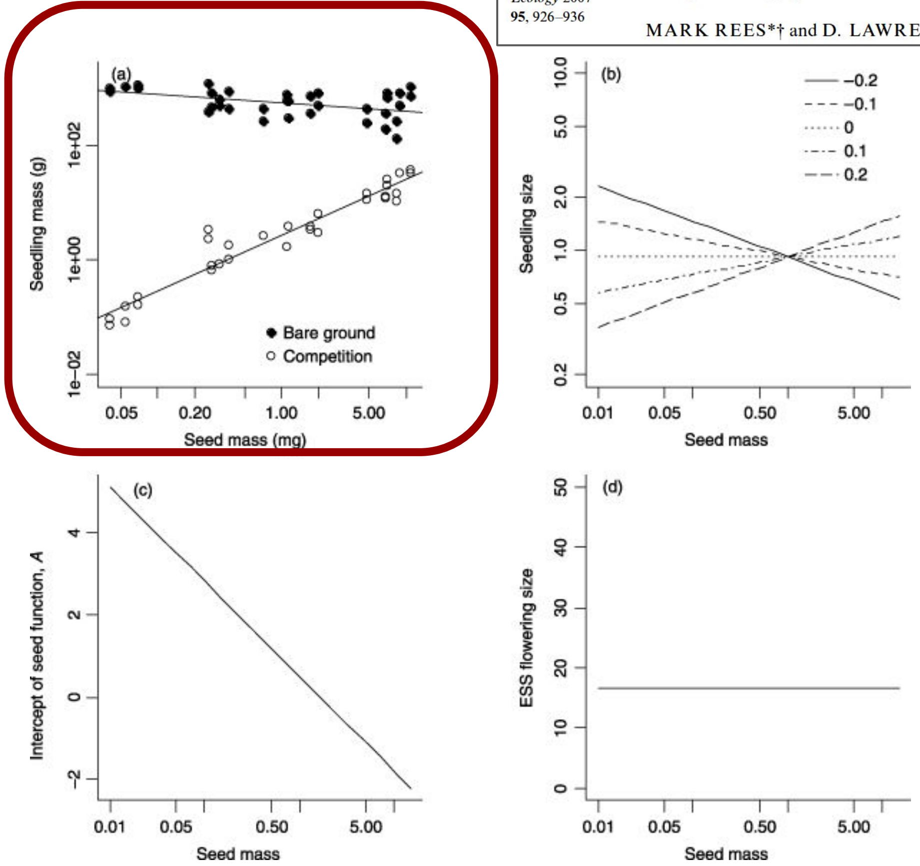
Journal of Ecology 2007 95, 926–936

Why do big plants make big seeds?

MARK REES*† and D. LAWRENCE VENABLE‡

Competition-Colonization and Seed Traits





Competition-Colonization: Evidence

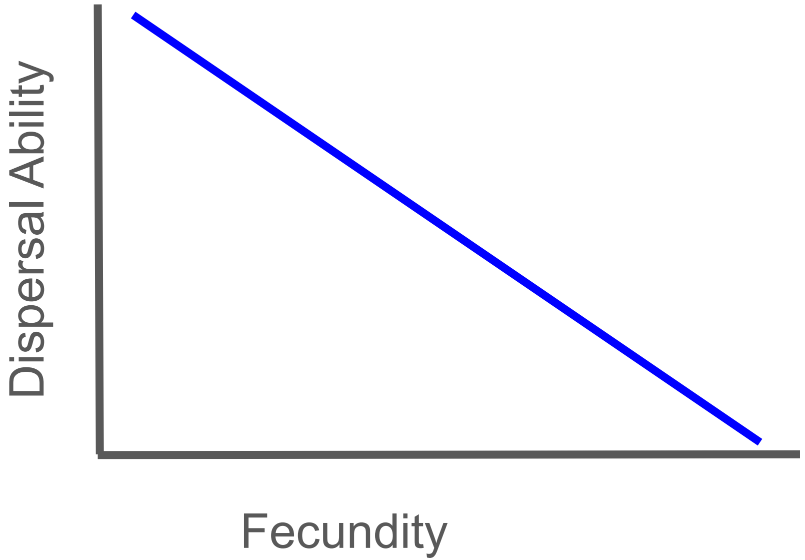
- Mixed evidence
- May be more important in some systems than others
- Interacts with other coexistence mechanisms

Patch Heterogeneity

- So far, we’ve assumed that patches are identical
- However, this isn’t going to be the case in reality
- Patch differences will also impact coexistence

Patch Heterogeneity - Distances

- Different distances between patches
- Coexistence via dispersal/fecundity trade-off



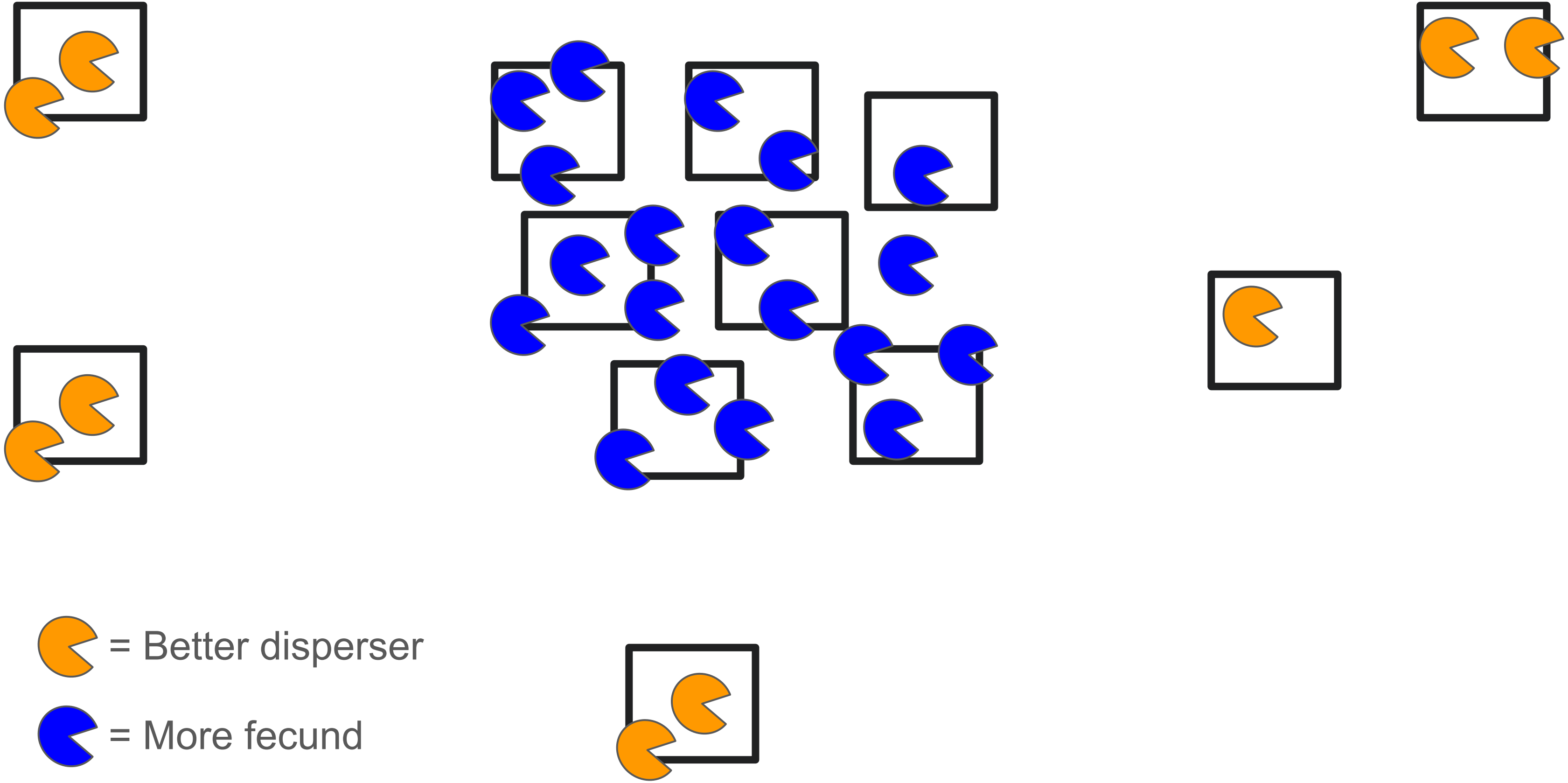
Patch Heterogeneity - Distances

Superior disperser: does better in low-density patches
Superior reproducer: does better in high-density patches

Superior colonizer species are concentrated in low-density patches
Superior competitor species are concentrated in high-density patches

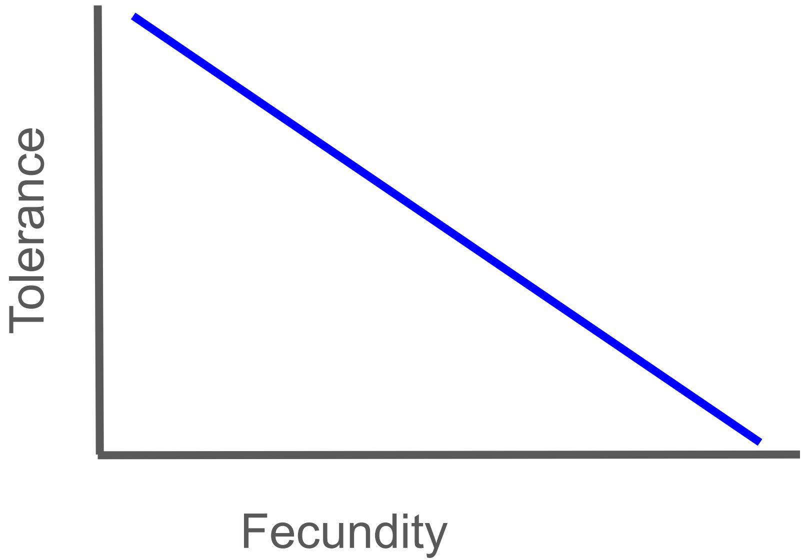
- Increases a_{ii} relative to a_{ji}

Patch Heterogeneity - Distances



Patch Heterogeneity - Quality

- Different patch qualities
- Coexistence via tolerance/fecundity trade-off



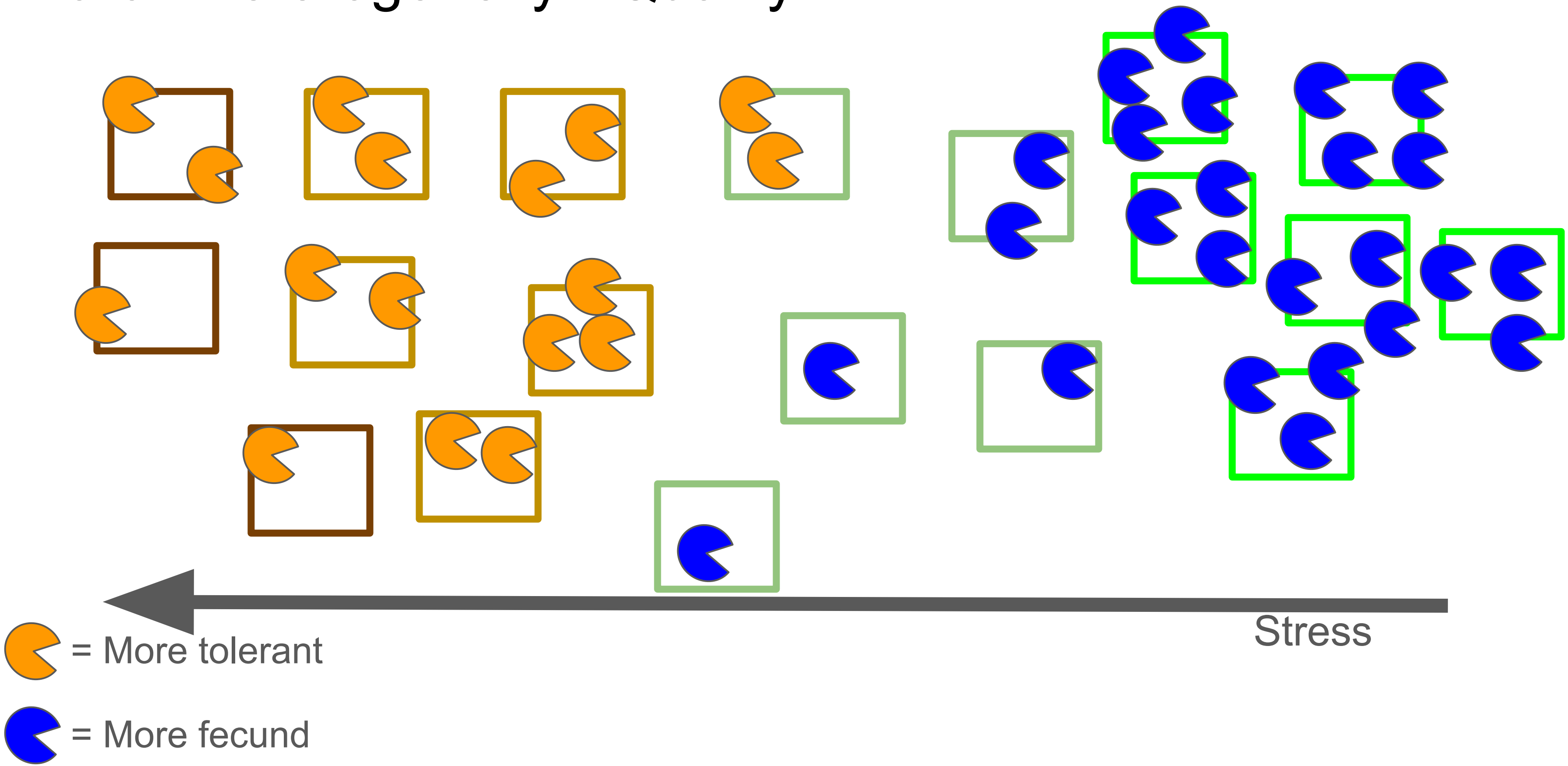
Patch Heterogeneity - Quality

Superior tolerator: does better in low-quality patches
Superior reproducer: does better in high-quality patches

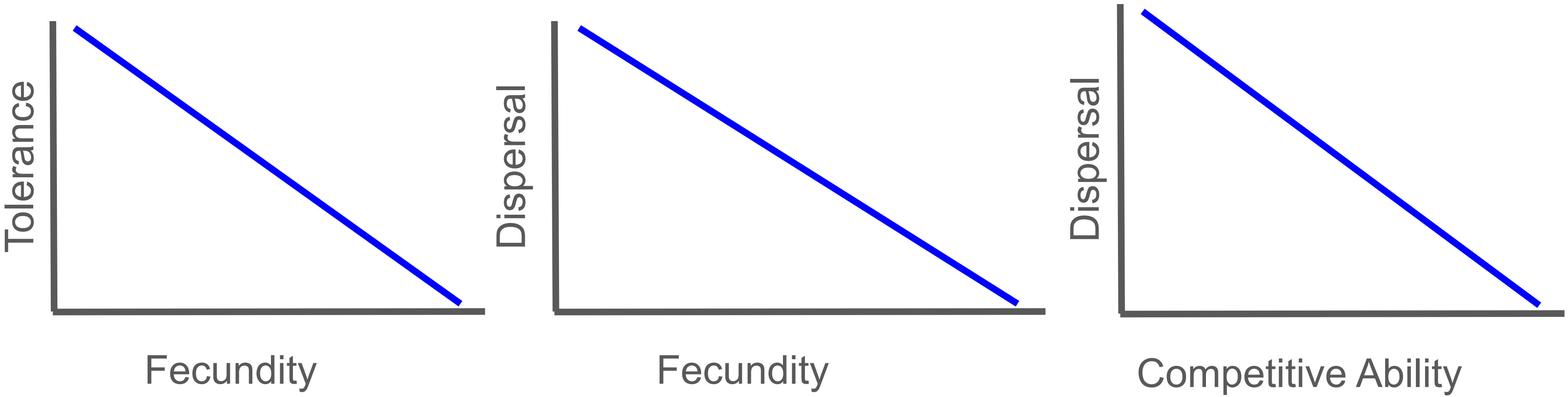
Superior tolerator species are concentrated in low-quality patches
Superior reproducer species are concentrated in high-quality patches

- Increases a_{ii} relative to a_{ji}

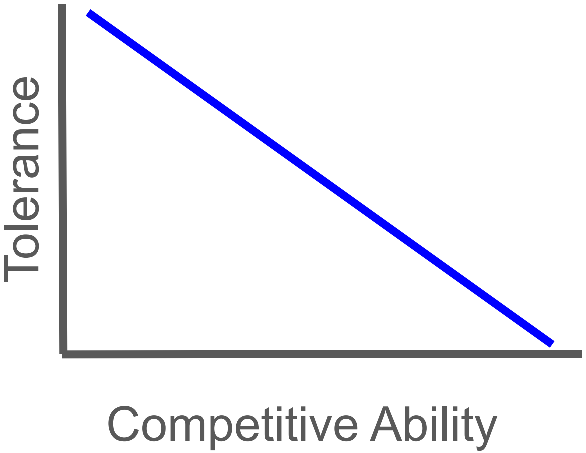
Patch Heterogeneity - Quality

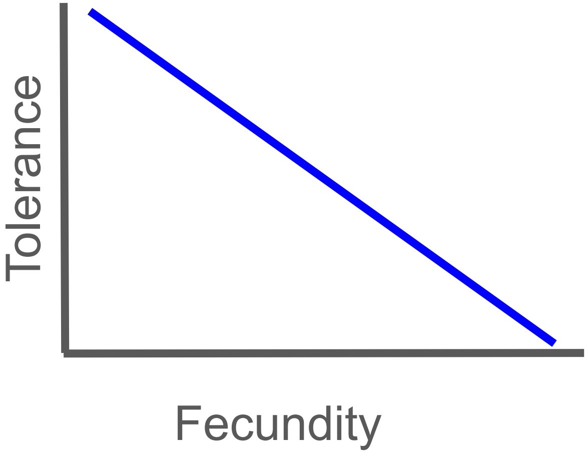


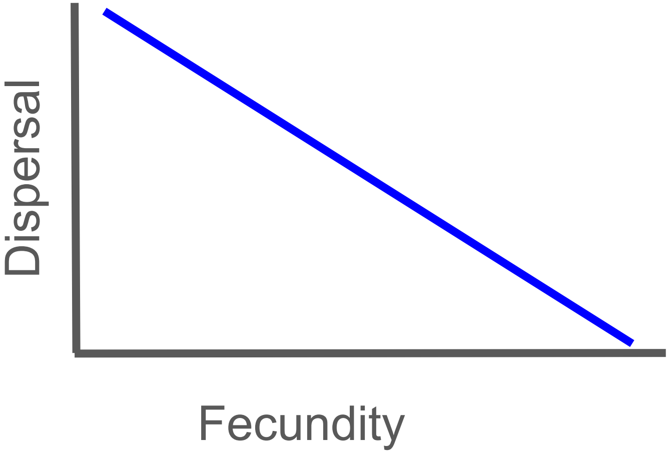
Patches and Coexistence

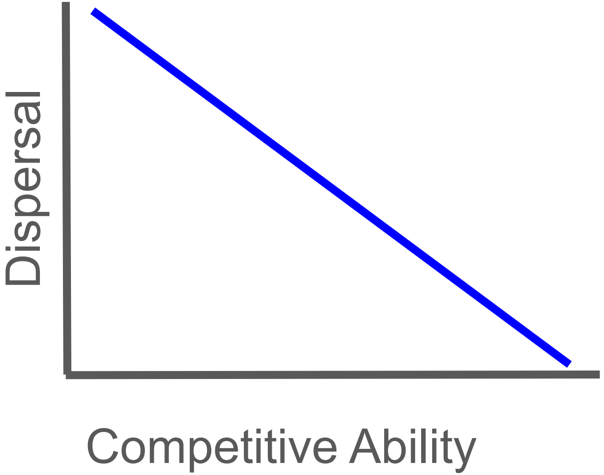


Patches and Coexistence



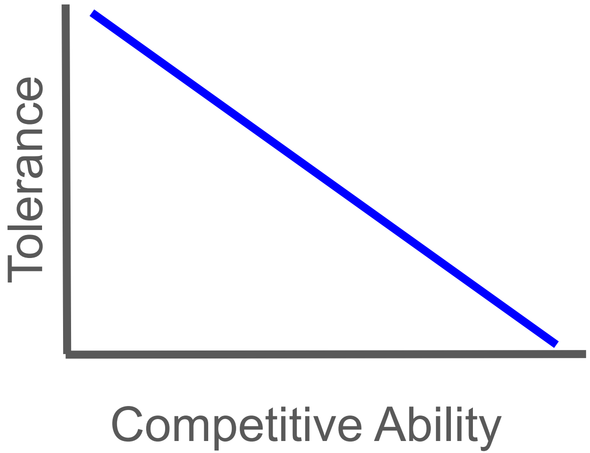


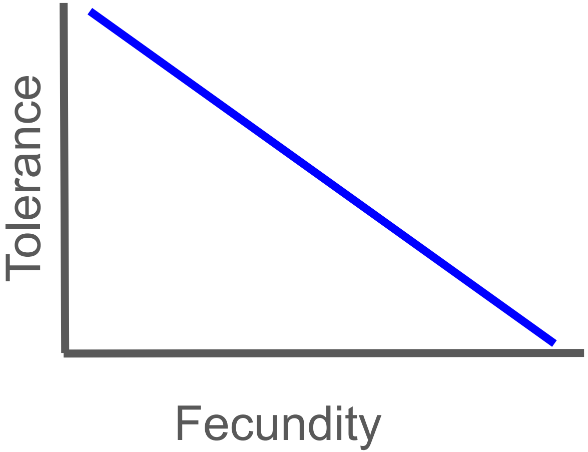


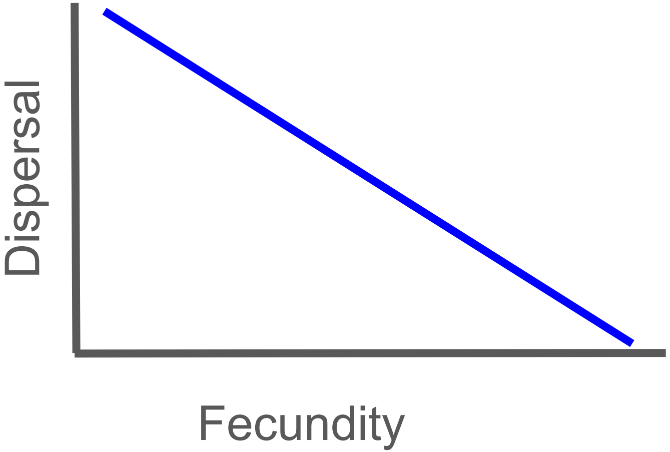


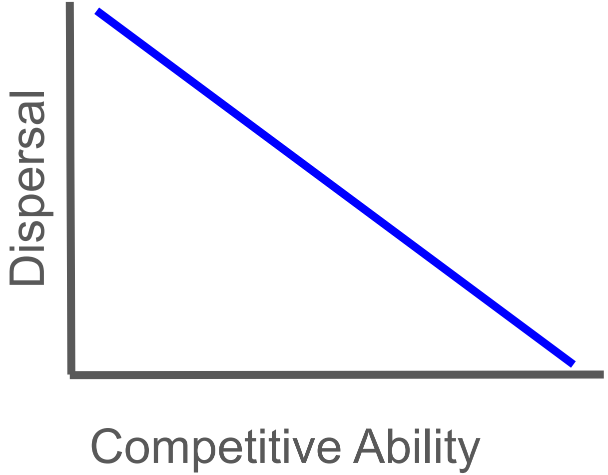
Patches and Coexistence

Common feature: trade-offs allow species to be concentrated in different patches.









Patches and conservation

These coexistence mechanisms may require disturbance to create empty patches

- Thus the disturbance regime is critical
- Too much OR too little disturbance can cause extinctions
- Complicates management decisions

Next class: Metacommunities